

Basics 39: Wave-Particle Resonance

V - Plasma waves and Alfvén physics

Phase synchronism, orbital frequencies, resonant energy exchange, and the origin of energetic-particle drive

Textbook draft, version 1.0. A plasma wave can exchange substantial energy with a charged particle only when the particle repeatedly encounters nearly the same phase of the wave. This chapter develops that statement from the wave phase, derives transit, bounce, drift, and cyclotron resonance conditions, explains the sign of energy exchange, and connects single-particle synchronism to the kinetic growth rate used in Module 5.

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1 Learning objectives

After completing this note, the reader should be able to:

- Explain resonance as phase synchronism rather than mere equality of two scalar frequencies.
- Derive a resonance condition by differentiating the wave phase along a particle orbit.
- Distinguish transit, bounce, drift, and cyclotron resonances.
- Relate the work $q\mathbf{E} \cdot \mathbf{v}$ to wave growth or damping.
- Explain why distribution-function gradients determine the net direction of energy transfer.
- Identify resonance broadening and nonlinear trapping scales.

2 Prerequisites and reading strategy

Recommended prerequisites are Basics 7, 13, 31, 32, 34, and 41. First follow the phase argument geometrically. Then reproduce the action-angle derivation and the reduced resonant-pendulum model.

3 The elementary phase-synchronism idea

Consider a one-dimensional electrostatic wave

$$E(x, t) = \text{Re}\left\{\tilde{E}e^{i(kx-\omega t)}\right\}. \quad (3.1)$$

A particle moving approximately as $x(t) = x_0 + vt$ samples the phase

$$\Phi(t) = kx(t) - \omega t = \Phi_0 + (kv - \omega)t. \quad (3.2)$$

If $kv - \omega$ is large, the particle rapidly alternates between accelerating and decelerating phases, and the long-time work nearly cancels. If

$$\boxed{\omega - kv \simeq 0}, \quad (3.3)$$

the phase changes slowly and coherent energy exchange can accumulate. Resonance is therefore a statement about a slowly varying phase.

4 Energy exchange with a wave

The particle kinetic energy obeys

$$\frac{d}{dt} \left(\frac{mv^2}{2} \right) = q\mathbf{E} \cdot \mathbf{v}. \quad (4.1)$$

For a sinusoidal wave, the sign of the instantaneous work depends on phase. A single resonant particle can gain or lose energy depending on its initial phase. A distribution of particles produces a systematic result only after phase averaging and accounting for how many particles occupy neighboring velocities or actions.

For a weak electrostatic wave, particles slightly slower than the phase velocity $v_\phi = \omega/k$ tend to be accelerated while particles slightly faster tend to be decelerated. A monotonically decreasing distribution has more slower particles than faster particles, so particles gain net energy and the wave damps. A positive slope in the resonant region can reverse the sign and drive the wave.

5 Toroidal mode phase

A perturbation in toroidal geometry is commonly expanded as

$$\tilde{F}(\psi) \exp[i(m\theta - n\phi - \omega t)], \quad (5.1)$$

where m and n are poloidal and toroidal mode numbers. Along an orbit,

$$\frac{d\Phi}{dt} = m\dot{\theta} - n\dot{\phi} - \omega. \quad (5.2)$$

For a simple passing orbit with mean frequencies ω_θ and ω_ϕ , phase synchronism requires

$$\boxed{\omega = m\omega_\theta - n\omega_\phi}. \quad (5.3)$$

Different sign conventions for the Fourier phase produce equivalent formulas with corresponding sign changes. The invariant content is $d\Phi/dt \simeq 0$.

6 Action-angle formulation

An integrable unperturbed orbit can be described by actions $\mathbf{J}$ and angles $\boldsymbol{\alpha}$, with

$$\dot{\boldsymbol{\alpha}} = \boldsymbol{\Omega}(\mathbf{J}), \quad \dot{\mathbf{J}} = 0. \quad (6.1)$$

A perturbation has Fourier components

$$H_1 = \sum_{\boldsymbol{\ell}} H_{\boldsymbol{\ell}}(\mathbf{J}) e^{i(\boldsymbol{\ell} \cdot \boldsymbol{\alpha} - \omega t)}. \quad (6.2)$$

The phase of one component evolves as

$$\dot{\Phi}_{\boldsymbol{\ell}} = \boldsymbol{\ell} \cdot \boldsymbol{\Omega} - \omega. \quad (6.3)$$

The general resonance condition is therefore

$$\boxed{\omega - \boldsymbol{\ell} \cdot \boldsymbol{\Omega}(\mathbf{J}) = 0}. \quad (6.4)$$

In toroidal energetic-particle notation this is often written schematically as

$$\boxed{\omega - n\omega_{\phi} - m\omega_{\theta} - \ell\omega_b - \omega_d \simeq 0}, \quad (6.5)$$

with the understanding that not every term is independent in every coordinate convention.

7 Transit, bounce, drift, and cyclotron resonances

7.1 Passing-particle transit resonance

Passing particles circulate around the torus. The dominant synchronism involves their parallel transit frequency and the field-aligned wave number:

$$\omega - k_{\parallel} v_{\parallel} \simeq 0. \quad (7.1)$$

This is the natural resonance for shear-Alfvén waves whose electric field and magnetic perturbation vary along field lines.

7.2 Trapped-particle bounce resonance

Trapped particles oscillate between mirror points with bounce frequency ω_b . Fourier analysis of the periodic bounce motion produces harmonics $\ell\omega_b$, giving

$$\omega - n\omega_\phi - \ell\omega_b \simeq 0. \quad (7.2)$$

The integer ℓ labels the bounce harmonic.

7.3 Precessional or drift resonance

The guiding center slowly precesses because of curvature and grad- B drifts. Low-frequency modes can resonate with the toroidal precession frequency:

$$\omega - n\omega_d \simeq 0. \quad (7.3)$$

This is especially important for trapped energetic particles and low-frequency energetic-particle modes.

7.4 Cyclotron resonance

When the wave frequency approaches harmonics of the gyrofrequency,

$$\omega - k_\parallel v_\parallel - \ell\Omega_c \simeq 0, \quad (7.4)$$

particles can exchange perpendicular energy with the wave. The low-frequency kinetic-MHD models used for Alfvén eigenmodes normally average over gyromotion and therefore retain cyclotron effects only through reduced finite-Larmor-radius corrections or closures.

8 Why gradients produce drive or damping

Linear kinetic theory produces resonant terms with a denominator

$$\frac{1}{\omega - \ell \cdot \mathbf{\Omega} + i0^+}. \quad (8.1)$$

Its imaginary part selects the resonant surface in phase space:

$$\text{Im} \frac{1}{x + i0^+} = -\pi\delta(x). \quad (8.2)$$

The power transfer contains a directional derivative of the equilibrium distribution,

$$P_{\text{res}} \propto - \int d\mathbf{J} \delta(\omega - \boldsymbol{\ell} \cdot \boldsymbol{\Omega}) \left(\boldsymbol{\ell} \cdot \frac{\partial f_0}{\partial \mathbf{J}} \right) |H_{\boldsymbol{\ell}}|^2. \quad (8.3)$$

Thus resonance identifies *where* exchange occurs, while the distribution gradient determines its net sign. Energetic-particle radial gradients, pitch anisotropy, and inverted energy-space populations can drive an Alfvén mode.

9 Finite resonance width and nonlinear trapping

A finite growth rate, collisions, orbit stochasticity, or finite observation time broadens the ideal delta-function resonance. Near one isolated resonance, a canonical transformation gives the pendulum Hamiltonian

$$H_{\text{res}} \simeq \frac{1}{2} G (\Delta J)^2 + h \cos \Phi. \quad (9.1)$$

Trapped resonant particles oscillate inside a phase-space island. The characteristic bounce or trapping frequency scales as

$$\omega_{\text{tr}} \sim \sqrt{|Gh|}. \quad (9.2)$$

When ω_{tr} becomes comparable to the linear growth rate, particle trapping can flatten the resonant gradient and saturate the mode.

10 Connection to the Kinetic MHD-PINN project

Module 4 supplies ω and the mode structure. Module 3 supplies orbit frequencies and trajectories. Module 2 supplies f_0 . Module 5 combines them to evaluate resonance locations and energetic-particle drive. A surrogate must preserve the sharp dependence on detuning and the sign dependence on distribution gradients; a classifier trained only on mode labels would discard the most important physics.

11 Computational laboratory

Goal: visualize phase locking and nonlinear trapping.

1. Integrate $\dot{\Phi} = G\Delta J$ and $\dot{\Delta J} = h \sin \Phi$ for several initial conditions.
2. Plot trajectories in $(\Phi, \Delta J)$ and identify passing and trapped regions.
3. Vary h and verify that island width scales approximately as $\sqrt{|h/G|}$.
4. Add weak decorrelation noise and measure resonance broadening.

5. Implement independently in Python and MATLAB with identical dimensionless parameters.

12 Summary

Wave-particle resonance is coherent phase synchronism along an orbit. The resonance condition is obtained by differentiating the perturbation phase along that orbit. Resonance alone does not determine stability: the distribution gradient and wave-particle coupling determine whether particles drive or damp the mode. Finite amplitude converts the linear resonance into a phase-space island and provides a natural saturation mechanism.

13 Exercises

1. Derive $\omega = kv$ from the one-dimensional wave phase.
2. Show how the toroidal resonance condition changes if the Fourier convention is $e^{i(n\phi - m\theta - \omega t)}$.
3. Explain why a flat distribution near resonance produces little net power transfer.
4. Derive the separatrix width of the pendulum Hamiltonian.
5. List the quantities Modules 2–5 must exchange to evaluate a resonance consistently.

Symbols and acronyms used in this document

Symbol / acronym	Name	Meaning in this document
Φ	Wave-particle phase	Perturbation phase evaluated along a particle orbit.
J	Action variables	Orbit invariants conjugate to periodic angles.
Ω	Orbit frequencies	Transit, bounce, precession, or gyro frequencies.
γ	Growth rate	Positive when wave amplitude increases under the adopted convention.
ω_{tr}	Trapping frequency	Nonlinear bounce frequency inside a resonant phase-space island.

14 References

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4. L. Chen and F. Zonca, “Physics of Alfvén waves and energetic particles in burning plasmas,” *Reviews of Modern Physics*, 2016.